



Unpacking the drivers of the socio-environmental sustainability of new ventures: insights from innovative digital start-ups in Italy

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Abstract

Thanks to the pervasive adoption of digital technologies across different industries, “digital-born companies” nowadays play a pivotal role in the current techno-economic scenario. The concurrent sustainability imperative has generated constraints and entrepreneurial opportunities for new ventures. A narrow part of the entrepreneurship literature has analyzed the relationship between the emerging digital phenomenon and the transformative capacity of ICT start-ups as a potential key driver to tackle grand societal challenges due to their ability to alter business models and/or optimize existing resources. Following this line of reasoning, we analyze how strategic orientation and boundary-spanning activities of ICT start-ups relate to their socio-environmental sustainability. Poisson family estimations (negative binomial and zero-inflated negative binomial) of 172 innovative ICT start-ups based in Italy reveal different driving factors of their socio-environmental sustainability. Our work provides novel insights into the sustainability drivers of digital start-ups, suggesting differentiated strategies for entrepreneurs aimed at fostering socio-environmental sustainability as one of their primary goals. Market strategic orientation is more associated with social sustainability, while technological strategic orientation is more associated with environmental sustainability. The development of a wide portfolio of partnerships seems beneficial only for social sustainability, with no effect on environmental sustainability.

Keywords Digital entrepreneurship · Green sustainability · Social sustainability · ICT start-ups · Boundary spanning · Strategic orientation

Introduction

In the past decade, the shift towards digital products and processes has significantly transformed economic systems and innovation activities (Yoo et al., 2012; Porter & Heppelmann, 2015). This emerging digital paradigm, characterized by the reproducibility, distributedness, and generativity of digital objects (Kallinikos et al., 2013), has spurred the global rise of ICT start-ups (Nambisan et al., 2019). These “digital-born companies” are pivotal in today’s techno-economic landscape, embodying cutting-edge knowledge, advanced tools, and frontier competencies (Von Briel et al., 2018; Zaheer et al., 2019; George et al., 2021; Mazzoni et al., 2021).

Concurrent with this first trend, the literature has stressed the burgeoning importance of sustainability as a novel driver for both the practice and scholarly investigation of entrepreneurship (George et al., 2021). This focus on sustainability is tied to the concept of a “*tipping point*” (Lenton et al., 2008), marking a phase of “*critical transition*” (Scheffer et al., 2012) where global challenges like the environmental crisis, energy transition, and resource shortages become central issues.

Despite this dual focus on digital innovation and sustainability, the entrepreneurship literature has sparsely explored the intersection between the digital phenomenon and the transformative potential of ICT start-ups (Elia et al., 2020; Gregori & Holzmann, 2020; George et al., 2021; Lammers et al., 2022; Böttcher et al., 2024). The increasing recognition of sustainability as a standalone objective for both established companies and new ventures, driven by a generational leadership shift, remains underexamined in digital businesses (Lammers et al., 2022), where financial and performance goals still dominate the debate (Lammers et al., 2022).

Digital technologies, as external enablers, can support various objectives (von Briel et al., 2018), including sustainability (Böttcher et al., 2023). Examples of this include digital-only products such as sustainable software solutions (e.g., micro-mobility services and platforms for social purposes), sustainable-product-service systems (e.g., smart meters to save energy), and business intelligence that utilizes big data for social environmental purposes, all of which illustrate the intersection of digital technology and sustainability (Böttcher et al., 2024). Given entrepreneurship’s catalyst role in generating non-economic social benefits (Volkman et al., 2021), digital businesses can drive sustainability through new entrepreneurial opportunities harnessing ICT’s generative potential (Zaheer et al., 2019; Mulligan & Kelly, 2021; Faludi, 2023). Following this line of reasoning, we underline the importance of analyzing the sustainability drivers of ICT start-ups. However, few studies have empirically examined the determinants of sustainability orientation in new digital ventures, primarily focusing on the environmental aspect (i.e. York & Venkataraman, 2010; Keskin et al., 2013; Dickel et al., 2018; Gu & Su, 2018; Hörisch et al., 2019). There is a lack of research on socio-environmental sustainability in new ventures and few studies that consider sustainability as a standalone goal for start-ups (Horne & Fichter, 2022).

Our paper aims to address these gaps by examining the relationship between key start-ups aspects and the distinctive social and environmental dimensions of digital start-ups’ sustainability. Specifically, we analyze the impact of strategic orientation and boundary spanning activities. These theoretical approaches are essential

yet understudied in identifying the drivers of socio-environmental sustainability in start-ups.

Strategic orientation encompasses the principles guiding a firm's actions across multiple dimensions (Gatignon & Xuereb, 1997; Hakala, 2011). Our focus is on market and technology strategic orientations due to their centrality to customer engagement and resource allocation in new ventures (Guo et al., 2020).

Boundary spanning refers to the extent to which a start-up leverages a network of information sources and collaborations to enhance its market and technology knowledge (Laursen & Salter, 2006; Lopez-Vega et al., 2016; Wang et al., 2020).

Our study is based on data from 172 digital start-ups in Italy (included in the national register for innovative new ventures), sourced from an original survey conducted in 2018. The decision to focus on Italy as the context for analysis stems from two primary considerations. First, Italy is a traditional manufacturing country, where ICT start-ups may act as complementors to solve existing business problems of incumbents (Capozza et al., 2018; Colombelli et al., 2023). Second, there is a central government commitment to fostering the birth and development of the most promising start-ups as vehicles to promote growth. This is also exemplified by the availability of a national register of innovative start-ups since 2012 (Colombelli et al., 2024).

This study makes two main contributions. First, it contributes to the extant entrepreneurship literature by connecting digital start-ups to the emerging sustainability literature, providing novel insights into the mechanisms driving social and environmental sustainability in new ventures.

Second, it integrates the strategic orientation and boundary spanning frameworks, which are typically examined separately and primarily in incumbent firms, to understand the internal and external factors crucial to the success of new firms.

The remainder of the paper is organized as follows: Section 2 outlines the theoretical background and hypotheses development; Section 3 details the methodological approach, including data collection, variable definition, and model estimation; Section 4 presents the results; Section 5 discusses the findings; and Section 6 concludes with the study's implications and limitations.

Theoretical background

Strategic orientation and socio-environmental sustainability

New firms have been identified as a very promising channel to address societal problems (Shane & Venkataraman, 2000; York & Venkataraman, 2010; Bianchi & Verganti, 2021; Volkmann et al., 2021).

Early phases of start-up growth are often driven by short-term business goals aimed at increasing their survival chances (Dickel et al., 2018; Hörisch et al., 2019). Nevertheless, the rising awareness of global socio-environmental issues in the entrepreneurship literature has shaped sustainability as a potential asset and source of new business opportunities (Anand et al., 2021; George et al., 2021).

This emphasis also extends to new digital ventures, which, thanks to the enabling capacities of digital tools (von Briel et al., 2018), have the potential to tackle socio-environmental challenges on a global scale (Zaheer et al., 2019; Böttcher et al., 2023, 2024). For digital start-ups, the challenge may lie in pursuing multiple avenues for growth, with sustainability not being the foremost priority. In this case, as shown for traditional manufacturing incumbents (see Denicolai et al., 2021), start-ups may struggle to achieve sustainability goals.

Considering this, the strategic orientation of new ventures is particularly important and can significantly affect how they contribute to socio-environmental sustainability. The strategic orientation of the firm refers to the principles that direct and influence its activities to conquer and maintain a durable competitive advantage (Gatignon & Xuereb, 1997). Strategic management literature has identified four main types of strategic orientation, distinguishing between market orientation, technology orientation, learning orientation, and entrepreneurial orientation (Hakala, 2011). In the realm of digital start-ups, the two most relevant dimensions are the market and technology orientations (Guo et al., 2020), which reflect the “market-pull” and “technology push” philosophies. While interactions between the two philosophies could occur (see Guo et al., 2020), our paper focuses on the predominant strategic orientation, as start-ups typically exhibit either market-driven or technology-driven business models.

We claim that the market orientation of start-ups affects social sustainability, while technology orientation influences environmental sustainability, and we provide support for this vision in the following.

Market orientation and social sustainability

Market orientation refers to the capacity of firms to deliver superior customer value by correctly framing customer targets, understanding latent user needs and behaviors, and eventually anticipating future emerging trends (the so-called market-pull philosophy) (Morgan & Strong, 1998; Noble et al., 2002; Chen et al., 2015; Jansson et al., 2017; Wang et al., 2019; Guo et al., 2020). In the literature, there are different approaches to market orientation; we align with the original conceptualization of Narvel and Slater (1990), who claim that a market orientation is composed of three behavioral components and two decision criteria – customer orientation, competitor orientation, inter-functional coordination, a long-term focus, and a profit objective. In addition, we also consider brand focus, as suggested by Noble et al. (2002), as a relevant component of the market orientation.

The detection and monitoring of emerging consumer values, for instance, those related to societal themes, is a distinctive strategy pursued by start-ups willing to strengthen their market positioning (Fraj-Andrés et al., 2009). The literature on social entrepreneurship explored the above-mentioned trend by focusing on firm societal commitment but generally disregarding the analysis of the role of market strategy in capturing customers’ attention (De Lange, 2017; He et al., 2022). Literature on corporate social responsibility is mainly concerned with the impact of established firms on society, searching for actions to improve the governance of the organizations and of the supply chain, rarely tackling issues related to market development (Retolaza et al., 2009). In this respect, recent studies have measured Eco-Corporate social respon-

sibility (ECO-CSR) managerial strategy and its influence on marketing and profit of firms (Babic et al., 2024). This literature does not focus on start-ups and their societal commitment, which is often not measurable through the usual ESG (environmental and social governance) indicators but can be captured only through qualitative analysis that enables one to discover the social impact of their business. Nevertheless, start-ups are particularly keen to enter the market and conquer customers' loyalty through distinctive products, signaling their uniqueness. This might positively affect their social sustainability commitment since changing lifestyles and consumption patterns are increasingly calling for more sustainable products, for which consumers are more likely to pay a price premium (Biswas & Roy, 2015). Consumers' perception of socially responsible products is among the key determinants of purchase decisions; therefore, to accomplish this demand for ethical products and to stand out from the competition, companies need to provide accurate product information and invest in social recognition (Straughan & Roberts, 1999; LaRoche et al., 2001). In a fierce competition arena, start-ups more engaged with their customers and with solid market knowledge are better equipped to perceive the expectations and feelings of their potential market and consequently to better understand and communicate the social value of their products and services (Dean & McMullen, 2007; Keskin et al., 2013; Wang et al., 2019; Ćorić et al., 2020). Considering also the advent of transversal shocks such as the pandemic, market awareness candidates as a promising antecedent of social commitment of start-ups that are increasingly perceiving societal engaged actions as part of the value they can deliver to customers (Sedita & Trovò, 2021; Liu et al., 2021). Based on this, we hypothesize what follows:

Hp1. Start-ups with a market strategic orientation reveal a stronger interest in social sustainability achievements.

Technological orientation and environmental sustainability

Technological orientation refers to the capability of firms to adopt a strategy focused on product development, usually with consistent investment in R&D and on-the-frontier competencies and the use of advanced technologies (the so-called technology-push) (Mu et al., 2017; Guo et al., 2020).

Start-ups are often seen as the cradle of technological development, which often intercepts the interests of well-developed companies, especially when looking for green solutions (Bendig et al., 2022). Large firms prefer to externalize this function and acquire ready-to-use green technologies, due to the lack of time to investigate too narrow technical problems (with uncertain results) or the absence of hyperspecialized competencies. Also, the proliferation of global problems (such as the environmental crisis), which are complex and systemic and involve organizational, market, and institutional aspects, transforms start-ups into a preferred locus for green innovation. This is because start-ups engaged with technological innovation combine the capacity of visionary entrepreneurs to understand the optimal solutions to address complex problems, with a very high level of cutting-edge technical skills (Gu & Su, 2018). When the technical capacity encounters an unmet problem, technology-intensive

new ventures set the competition over the frontier, with a process of creative destruction of the dominant solution or a significant improvement of the existing solutions (York & Venkataraman, 2010; Bianchi & Verganti, 2021).

The innovative culture of a start-up and the mindset of the entrepreneur, often quite prominent in the case of new technological ventures, are fundamental ingredients to apply (and translate) the abovementioned technical knowledge to alternative framing of existing problems or for imagining completely “unknown unknowns” scenarios (Lombardi & Mazzoni, 2023). In particular, the willingness to create new goods/services or the use of advanced technology to rethink or replace existing dirty production processes and fossil energy sources is a sign of the potential capacity of technology-oriented start-ups to promote the adoption of sustainable initiatives (Keskin et al., 2013; Bhupendra & Sangle, 2015; Gu & Su, 2018; Kuzma et al., 2020).

Besides, the technological orientation of start-ups is not only bound to the creation of high-tech solutions to mitigate climate change and carbon emissions but also to the adoption of green technologies or cleaner organizational behaviors in terms of energy consumption and waste management. Tech-friendly start-ups may own the absorptive capacity (Cohen & Levinthal, 1990) required to spot opportunities for the use and implementation of tools and methods to reduce their carbon footprint and the overall environmental impact. Following the evolutionary approach to the study of entrepreneurship (Breslin, 2008), technological orientation is configured as an ability that allows one to identify opportunities (variation), develop strategies (selection), and replicate successful routines (retention), thus increasing the probability to adopt environmentally sustainable behaviors.

Based on this, our hypothesis is the following:

Hp2. Start-ups with a technological strategic orientation reveal a stronger interest in environmental sustainability achievements.

Boundary spanning and socio-environmental sustainability

Boundary spanning is the process aimed at defining internal corporate borders and eventual trajectories of expansion (Powell et al., 1996; Rosenkopf & Nerkar, 2001; Schotter et al., 2017). Boundary spanning has become an increasingly complex activity, because of the difficulty of coordinating a variety of external knowledge flows such as suppliers, customers, universities, and occasionally competitors (Greco et al., 2016; Lopez-Vega et al., 2016; González-Moreno et al., 2019; Wang et al., 2020). Nevertheless, it is crucial for start-ups, which suffer from the liability of newness and smallness (Stinchcombe, 1965; Freeman et al., 1983), to identify an adequate portfolio of collaborations useful to grant access to tangible and intangible resources not internally available. These new resources are useful to better organize the business and to achieve a return on the investment in external actors, in terms of economic and innovation performance (Sedita & Apa, 2016; Spender et al., 2017; Gu & Su, 2018; Riandita et al., 2019; Guo et al., 2023). In particular, for digital start-ups, collaborating with external partners may favor win-win scenarios, where their cutting-edge competencies may be applied to incumbents’ business problems.

An increasing number of researchers are supporting the claim that boundary-spanning influences socio-environmental sustainability, and it is an effective way to support the increasing external stakeholder pressures on the topic (Testa et al., 2016; Dickel et al., 2018; Jesus & Jugend, 2021; Wang et al., 2020, Wang et al., 2022; Allal-Chérif et al., 2023).

As for the case of strategic orientation, we can distinguish between market and technological boundary-spanning activities (Guo et al., 2023). We claim that market boundary-spanning activities of start-ups affect social sustainability, while technological boundary-spanning ones influence environmental sustainability. In particular, we take into consideration two dimensions of boundary spanning, which are the breadth and the depth of the external knowledge search. Following a consolidated stream of literature on the topic (starting from the seminal work by Laursen & Salter, 2006), we explore the relative impact of the variety of knowledge sources (breadth) and the intensity of use of the knowledge source (depth). The two dimensions capture the exploration and exploitation of new external knowledge. Both dimensions are considered to influence the socio-economic sustainability of start-ups, as described in the following sections.

Market-related boundary spanning and social sustainability

A wide network of collaborations for market knowledge sharing allows start-ups to explore different contexts of formal and informal norms, learn best practices, and increase their legitimacy among partners (Rauter et al., 2019). This can help to promote the visibility of the start-up in entrepreneurial communities and to connect to key stakeholders that enable it to better frame its products and services to accomplish social sustainability goals. Not only the breadth of the collaboration for market knowledge sharing but also the depth of the collaboration with specific partners might affect the social sustainability of firms since some stakeholders might be more prone to support social initiatives or to be more active in the field (Holmes & Smart, 2009; Seuring & Gold, 2013). For instance, long-term projects with universities or well-routed collaborations with final customers requesting social responsibility can be an even more promising vehicle for promoting social commitment than the exploration of a wide range of collaborations.

Based on this, we hypothesize that start-ups that perform a market-related boundary spanning reveal a stronger interest in social sustainability achievements, in particular:

Hp3a: The breadth of external market knowledge search is positively related to start-ups' social sustainability achievements.

Hp3b: The depth of external market knowledge search is positively related to start-ups' social sustainability achievements.

Technological boundary spanning and environmental sustainability

Concerning the sharing of technical information, the effect of partnerships has been identified as a major driver to cope with changing regulations and standards, but also to increase organizational and production efficiency. The riskiness and uncertainty of complex projects can be mediated by the capacity of start-ups to access high-level technical knowledge. For instance, the possibility to prototype or conduct experiments exploiting intense collaborations with organizations holding complementary resources and competencies could favor the adoption of environmentally sustainable practices, as risks are shared among collaborators and a common set of capabilities are assembled to respond to complex challenges (Michelino et al., 2019; Wang et al., 2020). Moreover, the transition towards greener production models can be facilitated by deep relationships with key partners, who can find it more convenient to share key and very valuable technical information to obtain a win-win scenario for both (e.g. an increase in productivity) (Gu & Su, 2018; González-Moreno et al., 2019). In this regard, exposure to different stakeholders' influence can positively affect the problem framing of entrepreneurs and start-up managers, which can exploit and integrate external technical knowledge within the start-up. Chen et al. (2015) refer to the result of this intense collaboration with the term "eco-effectiveness". Moreover, also in this case, some specific collaborations might be more fruitful than others in terms of environmental sustainability, giving rise to a selection issue.

Based on this, we hypothesize that start-ups that perform a technological-related boundary spanning reveal a stronger interest in environmental sustainability achievements, in particular:

Hp4a: The breadth of external technological knowledge search is positively related to start-ups' environmental sustainability achievements.

Hp4b: The depth of external technological knowledge search is positively related to start-ups' environmental sustainability achievements.

Methodology

Data

To conduct the study, we rely on the data from the Italian Innovative Start-ups Register, which collects information on start-ups from 2012. Italy is an interesting testbed for our research purpose, for the presence of a national law that certifies the innovativeness of start-ups according to a set of established criteria¹. Additionally, as a

¹ According to article 25 of Decree-Law no. 179/2012, the definition of Innovative Start-up has to meet the following criteria: new company or incorporated for less than five years; headquarter in Italy; annual turnover lower than €5 billion; no profits' distribution; mission innovative oriented; no result of split-up or a company merger; able to satisfy at least one of the following innovation indicators: (1) expenses in R&D and innovation are at least 15% of the yearly costs; (2) 1/3 of employees with a Ph.D. or 2/3 with a master's degree are the holder, depositary, or licensee of a registered patent or software.

traditional manufacturing nation, Italy holds a keen interest in the potential of new digital start-ups to revitalize mature industries (Mazzoni & Innocenti, 2024).

The register in the year 2018 counted 2914 ICT start-ups (NACE sector 62 and 63), born between 2012 and 2017. With quota sampling, we stratified the entire population into mutually exhaustive subgroups, by geographical area. After, we evaluated the proportion in which the subgroups exist in the population, and we maintained this proportion in the sample. We scanned the database, searching for firms with active contacts. Out of 1104 contacted firms, we received replies from 179 start-ups (16% response rate), representative of the population of innovative ICT start-ups operating in Italy (see Table 7 with geographical coverage in the [appendix](#)). We then matched the sample with data coming from the AIDA² repository to retrieve information about the number of employees, obtaining a final sample of 172 start-ups. For a complete overview of the data collection process, see Fig. 1.

Most start-ups in the sample develop advanced solutions for 4.0 manufacturing, digital marketing, and e-commerce; a smaller portion offers digital solutions for cultural heritage and tourism as well as IoT services for smart living³.

We collected survey data from the administration of a questionnaire through telephone interviews conducted by a Computer-Assisted Telephone Interview (CATI) service supply company between January 8, 2018, and March 14, 2018⁴. The questionnaire is composed of 24 items divided into three parts and was addressed to one of the founders, reporting data and information on behalf of the other founders (when multiple founders existed). The first section collects data about the company's principal characteristics in terms of core business and structural indicators. The second section refers to the company strategy, including information about the strategic orientation and the collaborations for market and technology knowledge acquisition. The third section explores the firm's sustainability orientation, disentangling environmental and social aspects.

Variables

Dependent variables

Environmental sustainability (“*ENV sustainability*”) is the start-up commitment to the environment, measured through responses to the question: “What is the company's commitment to the environment?” The question includes 6 items with positive/negative options (see [appendix](#)), and the variable is measured as the total sum of the positive responses.

Social sustainability (“*SOC sustainability*”) is the start-up commitment towards the society/community of reference, and it is measured through responses to the

² Acronym for “Analisi informatizzata delle aziende italiane”.

³ for a similar approach see Colombelli et al. (2023).

⁴ This study acknowledges the University of Padova Research Project 2014 (PRAT 2014) called “Moving knowledge into action: exploring the micro-foundation of an innovation ecosystem.”



Fig. 1 Data collection process

question: “What is the company’s commitment to the community of reference?”. The question includes 6 items with positive/negative options (see Table 8 in the [appendix](#)), and the variable is measured as the total sum of the positive responses.

The choice of building our dependent variable summing up the six dummy variables is due to the assumption that the more initiatives are undertaken by the start-up, the greater the overall propensity towards socio and environmental sustainability, the main aim of the study⁵.

Independent variables

We measure the *strategic orientation* of start-ups using the question of the survey related to the strategic goals of the company, which is articulated across 6 items (see Table 9 in the [appendix](#)). To assess the main strategic orientation, we perform a factor analysis of the 6 items. For the reliability of the scale, we use Bartlett’s test and Kaiser–Meyer–Olkin (KMO) to test sampling adequacy and Cronbach’s Alpha. Bartlett’s test (χ^2 : 168.904; Degrees of freedom: 15; p-value: 0.000) suggests rejecting the null hypothesis that variables are not intercorrelated (see also correlation Table 10 in the [appendix](#)). The Kaiser–Meyer–Olkin (KMO) exhibits a value of 0.738, well above the threshold of 0.5, confirming the validity of the analysis.

Considering possible correlation among items, we use a promax oblique rotation (as in Apa & Sedita, 2017), generally preferred to orthogonal methods, for its better adherence to real data structures (Fabrigar et al., 1999). According to the examination of the eigenvalues and the screeplot (as suggested in Fabrigar et al.,

⁵ To test the validity and reliability of the summated scale, we conducted separate analyses on the 6 environmental and 6 social sustainability items in the raw data. We perform the Kaiser-Meyer-Olkin Measure of Sampling Adequacy (KMO), obtaining 0.713 and 0.784, respectively, for environmental and social sustainability, well above the threshold of 0.5. We also perform the Bartlett’s test obtaining (χ^2 : 139.768; Degrees of freedom: 15; p-value: 0.000) for environmental sustainability and (χ^2 : 200.430; Degrees of freedom: 15; p-value: 0.000) for social sustainability. The results suggest rejecting the null hypothesis that variables are not intercorrelated. Then, to validate the internal reliability of our dependent variables, we compute the Cronbach Alpha, obtaining 0.68 and 0.72, respectively, for environmental and social sustainability, well aligned with the suggested threshold of ≈ 0.60 – 0.70 for emerging themes that require further empirical consensus (Hair et al., 2014). As a further check for internal validity, we also used tetrachoric correlation, particularly indicated for dichotomous variables. We obtained that the 6 items included within social sustainability have a significant level of correlation in 80% of the total cases with an average value of 0.57; while the 6 items included within environmental sustainability have a significant level of correlation in 87% of the total cases with an average value of 0.50.

Table 1 Factor analysis

Factor	Eigenvalue	Difference	Proportion	Cumulative
Factor1	1.468	0.333	0.367	0.367
Factor2	1.136	0.361	0.284	0.651
Factor3	0.774	0.153	0.194	0.845
Factor4	0.622	.	0.155	1.000

LR test: independent vs. saturated: $\chi^2(6)=37.55$ Prob> $\chi^2=0.0000$
 Factor loadings (pattern matrix) and unique variances

Variable	Factor1	Factor2	Uniqueness
stra_revenues	0.555	0.574	0.363
stra_clients	0.589	0.546	0.356
stra_prodev	0.662	-0.467	0.344
stra_tech	0.613	-0.539	0.333

1999 and Hair et al., 2014) (see Table 11 and Figure 3 in the appendix), the factor analysis identified two factors to be retained. However, after factor rotation, two out of the six factors had significant cross-loadings, specifically the factors “stra_brand” and “stra_comm” (see Table 12). As recommended by Hair et al. (2014), to address this issue, we attempted different oblique rotation methods (e.g. e.g., promin, orthoblique). Nevertheless, cross-loading on the same items persisted, and as suggested by Hair et al. (2014), “stra_brand” and “stra_comm” become candidates to be deleted. Since the aim of this factor analysis is data summarization and not the analysis of specific items, we removed the two abovementioned variables. We removed one variable at a time according to their level of cross-loading to evaluate its persistence. However, even in the five-item model, the cross-loading persisted, forcing us to delete both problematic items, ending up with four items. Afterward, we run the scree plot and factor analysis again with the promax oblique technique, and we obtain the most efficient solution with eigenvalues bigger than 1 in the two-factor solutions (see Table 1)⁶.

To ensure better interpretability of the results, we followed Hair et al.’s (2014) recommendation and used the rotated factors as our new variables, as presented in Table 2⁷. We interpret factor 1 as the technological strategic orientation of start-ups

⁶ The same solution is reached with different rotation methods, namely: oblique oblimin and orthogonal quartimax. To control for the importance of the two deleted items, we run a correlation with the rotated factor variables built with all six items and the one made by four items. Original factor 1 was correlated at 91% with the new factor 1, and original factor 3 was correlated at 92% with the new factor 2, meaning that we are still able to capture data variability with such a solution.

⁷ To double-check our results, we also created a summated scale from the two-factor solutions, and the results are identical (variables created with unrotated factors and summated scales are correlated at 0.99%). To confirm the scale’s validity of our factor scores, we rely on the very high level of factor loadings (more than 0.80 for all four items) (see Guo et al., 2020). Then we report for the technological, strategic orientation Average Variance Extracted (AVE): 0.659 and Composite Reliability (CR): 0.795, and for the market strategic orientation AVE: 0.641 and CR: 0.781. These results are well above the recommended thresholds and in line with the existing literature on strategic orientation (Guo et al., 2020). Considering that our two-factor solution is based only on two dichotomous items each, Cronbach Alpha is not the most adequate measure in this case. Instead, we rely on the tetrachoric correlation between our four dichotomous items (two for each factor solution). Our findings indicate a noteworthy correlation, with internal items for market strategic orientation and technological orientation showing respective correlations of 0.59 and 0.48; while no significant correlation was observed among the other items.

Table 2 Factor analysis- rotated factors – strategic orientation of firms

Variable	Label	Factor 1 “Technology push”	Factor 2 “Market pull”	Uniqueness
stra_revenues	rapid growth in terms of revenues	-0.023	0.801	0.363
stra_clients	rapid growth in terms of the number of clients	0.021	0.800	0.356
stra_prodev	product/service development	0.803	0.045	0.344
stra_tech	technology development	0.821	-0.043	0.333

Note: In bold factor loadings higher than 0.5

(“*Technology-Push*”), given that product/service development and technology development are very often jointly pursued by new firms. We interpret factor 2 as the market strategic orientation (“*Market-Pull*”), considering that the most relevant variables relate to increased revenues and customer base, all actions devoted to the early market strategy of new ventures.

We measure collaboration strategy adopting the notion of breadth and depth, firstly operationalized by Laursen and Salter (2006) in the context of open innovation and then adopted by other researchers about digital entrepreneurship (see Guo et al., 2023) and green innovation (see González-Moreno et al., 2019; Wang et al., 2019). In comparison to the measures employed by Laursen and Salter (2006), who operationalize knowledge sources to build breadth and depth across a four-tier scale (“not-used”, “low”, “medium”, “high”), this study embraces a more refined adaptation of the metric, articulated through a seven-point Likert scale. This modification enhances the granularity and precision of the assessment, in alignment with the framework proposed by Chen et al. (2011).

Breadth measures the degree of diversity of collaborations and external search. In our case, we identify eight different typologies of collaborations (suppliers, clients, start-ups in the same sectors, start-ups in other sectors, companies in the same sector, companies in other sectors, private consultants, universities, and research institutions), creating a scale variable accounting for the number of different relationships owned by the start-up. Depth measures the intensity of the relationship. It is constructed using the same eight typologies of collaboration used in constructing breadth. In this case, each of the eight typologies is coded with 1 when the start-up reports that it intensively relies on this collaboration (corresponding to the two highest values of our seven-level Likert scale, which goes from 1 “no collaboration” to 7 “strong collaboration”). As in the case of breadth, we computed depth, summing up the values. Considering that our survey included a separate score for market information and technical information, we decided to compute breadth and depth for the two categories, as they may reveal different impacts on social and environmental orientation (for an overview of the questions, see Table 9). We, therefore, distinguished between “*Market Breadth*”, “*Technical Breadth*”, “*Market Depth*”, and “*Technical Depth*”.

Control variables

Some control variables have been added to the model. First, we add the number of founders (“*N_founders*”), since a larger founding team can face some coordination problems in setting strategic priorities, causing possible misalignment of incentives.

Second, we adopt a dummy variable to measure the break-even point (“*BEP*”) to control for the possible influence of a healthy accounting balance in supporting more sustainable attitudes of the company. Third, we control for the start-up dimension (“*Start-up size*”), which indicates the number of employees. Start-ups with more employees may have dedicated departments or at least people responsible for environmental and social aspects. Fourth, we control for the possible impact of gender diversity, as a determinant of ecological and social consciousness (Park et al., 2012). In this respect, we measure the percentage of women in the total number of employees (“*Women*”).

Tables 3 and 4 report the descriptive statistics and the correlation matrix. Concerning the latter, we register the highest level of correlation ($>0,5$) between market and technological depth (0,752) and between market and technological breadth (0,723). However, these couple of variables are tested in separate models, without risk of multicollinearity, as displayed in the results section (Section 4).

Model estimation

Given that our two dependent variables, social and environmental sustainability, are count variables ranging from “0” (indicating no actions towards the betterment of the environment and society) to “6” (indicating all possible actions towards the betterment of the environment and society included in the survey’s question), the most appropriate regression model falls under the Poisson family. Since the data shows overdispersion and a large proportion of zeros in both dependent variables (respectively 40% and 38% for social and environmental sustainability), we conducted tests on negative binomial regression (NB) and zero-inflated negative binomial regression (ZINB). Although ZINB models appear better suited to handling data overdispersion and a significant presence of zeros, we applied the corrected Vuong test to choose between NB and

Table 3 Descriptive statistics

Variable	Obs	Mean	Std. Dev.	Min	Max
SOC sustainability	172	1.538	1.686	0	6
ENV sustainability	172	1.48	1.512	0	6
Market Breadth	172	5.37	1.917	0	8
Market Depth	172	1.566	1.522	0	8
Technical Breadth	172	5.127	2.349	0	8
Technical Depth	172	1.561	1.633	0	8
Market-Pull	172	0	1	-2.671	0.672
Technology-push	172	0	1	-2.784	0.647
Women	172	0.139	0.251	0	1
BEP	172	0.457	0.5	0	1
N_founders	172	3.087	1.91	1	14
Start-up size	172	3.733	3.089	1	25

Table 4 Correlation matrix

Variables	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)	(12)
(1) SOC sustainability	1.000											
(2) ENV sustainability	0.205	1.000										
(3) Market Breadth	0.273	0.062	1.000									
(4) Market Depth	0.235	0.108	0.424	1.000								
(5) Technical Breadth	0.273	0.099	0.723	0.335	1.000							
(6) Technical Depth	0.279	0.160	0.369	0.752	0.462	1.000						
(7) Market-Pull	0.126	0.094	0.045	-0.059	0.116	0.049	1.000					
(8) Technology-push	0.119	0.186	0.195	0.084	0.188	0.140	0.131	1.000				
(9) Women	0.189	0.059	0.085	0.079	0.030	0.021	0.020	-0.058	1.000			
(10) BEP	0.075	0.084	0.125	0.069	0.212	-0.029	-0.182	0.089	-0.069	1.000		
(11) N_founders	0.025	-0.031	0.102	0.139	-0.026	0.167	0.038	0.060	0.091	-0.053	1.000	
(12) Start-up size	-0.025	0.117	0.071	0.146	-0.030	0.118	-0.052	0.058	-0.004	0.049	0.398	1.000

ZINB models (Desmarais & Harden, 2013; Blevins et al., 2015). In particular, Blevins et al. (2015) highlight that while ZINB may be better suited than NB when values are zero, this is not automatically true for higher values. In addition, it is important for ZINB to theoretically frame the generative process behind the zeros⁸.

Based on the results of the corrected Vuong Test, we found that for five out of six models with social sustainability as the dependent variable, NB was preferred. For all models with environmental sustainability as the dependent variable, ZINB obtained better scores, emphasizing the need to evaluate each case carefully, as recommended by the step-by-step procedure outlined in Blevins et al. (2015).

Results

The estimations, presented in Tables 5 and 6, followed a stepwise approach.

Table 5 shows the results when social sustainability is the dependent variable. We reported the results of the preferred model between NB and ZINB, following the test explained in Section 3.3.

In column 1, we first estimated the model only with control variables, finding that gender diversity among employees (% of women) is significantly correlated with the start-up's social sustainability. This result holds for all the models run in Table 5. We have also observed a positive and significant impact of BEP, which remains consistently significant in all ZINB versions (refer to Table 13 for the other estimations). However, we are unable to confirm the strength of this correlation across all model estimations, as the preferred models in most cases are NB, where BEP does not demonstrate any effect.

Column 2 shows the impact of the strategic orientation of firms. The model reports a significant and positive effect of the market-pull factor, which shows a higher magnitude of the coefficient if compared to that of the technology-push, thus confirming Hp 1. The influence of the market-pull factor is persistently positive also in the specification reported in column 5, when market boundary spanning is simultaneously considered.

Column 3 shows a positive and significant effect of the breadth of market boundary spanning activities of the start-up on social sustainability, confirming Hp 3.a, and this result also holds in Column 5, when it is considered together with the market-pull strategic orientation. Hp3.b, which states the positive role of the depth of market boundary spanning activities is not confirmed in all models' specifications, suggesting more caution in assessing the relationship between intense collaborations and social sustainability.

Column 4 shows a positive effect both of breadth and depth of technical boundary spanning on social sustainability, suggesting that not only collaborations for market

⁸ ZINB regression models consist of two components. The first component is responsible for estimating count dependent variables like a NB regression. The second component employs a logit model to elucidate the causes behind the zeros of the dependent variable. Therefore, it is crucial to develop models that accurately consider the underlying mechanisms that generate these zeros (Desmarais & Harden, 2013). It is worth noting that zeros may follow a distinct rationale from the positive values, and the research context and data employed can help the researcher to decide if two different models are needed.

Table 5 Determinants of social sustainability

	ZINB(1)	NB(2)	NB(3)	NB(4)	NB(5)	ZINB(6)
Women	0.492* (0.197)	0.812** (0.274)	0.566* (0.249)	0.670** (0.255)	0.582* (0.253)	0.523* (0.214)
BEP	0.317* (0.151)	0.230 (0.169)	0.138 (0.165)	0.127 (0.166)	0.196 (0.166)	0.307* (0.153)
N_founders	-0.0244 (0.0519)	0.0226 (0.0486)	0.00630 (0.0416)	0.00952 (0.0425)	0.00484 (0.0419)	-0.0392 (0.0553)
Start-up size	0.0457 (0.0412)	-0.0175 (0.0374)	-0.0319 (0.0355)	-0.0264 (0.0413)	-0.0289 (0.0341)	0.0384 (0.0587)
Market-pull		0.173^Δ (0.0913)			0.177* (0.0879)	
Technology-push		0.152^Δ (0.0818)				0.0957 (0.0647)
Market Breadth			0.129* (0.0514)		0.121* (0.0527)	
Market Depth			0.0795 (0.0500)		0.0843^Δ (0.0499)	
Technical Breadth				0.0916* (0.0430)		0.0766* (0.0364)
Technical Depth				0.0974* (0.0439)		0.0549 (0.0387)
Constant	0.492** (0.163)	0.153 (0.186)	-0.505 (0.312)	-0.356 (0.271)	-0.517 ^Δ (0.312)	-0.0251 (0.274)
inflate bep	0.374 (0.421)					0.653 (0.523)
Constant	-0.933** (0.328)					-1.281** (0.454)
lnalpha						
Constant	-50.81*** (0.124)	-0.381 (0.238)	-0.521 ^Δ (0.276)	-0.582* (0.297)	-0.575* (0.281)	-77.35*** (0.103)
Observations	172	172	172	172	172	172
Pseudo R ²		0.020	0.031	0.034	0.036	
AIC	581.4	587.2	581.2	579.6	579.9	572.2
BIC	606.6	612.4	606.4	604.8	608.2	606.8
Chi2	10.35	19.02	23.37	23.06	32.44	32.67
Log-likelihood	-282.7	-285.6	-282.6	-281.8	-280.9	-275.1
LR test		36.99***	28.80***	24.67***	25.81***	
Vuong- AIC	58.23***					28.76***
Vuong- BIC	56.26***					27.77***
Max VIF	1.21	1.21	1.25	1.39	1.26	1.41

Standard errors in parentheses ^Δ $p < 0.1$, * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

Table 6 Determinants of environmental sustainability

	ZINB(1)	ZINB (2)	ZINB (3)	ZINB (4)	ZINB (5)	ZINB (6)
Women	0.273 (0.214)	0.240 (0.236)	0.236 (0.220)	0.244 (0.223)	0.186 (0.226)	0.280 (0.235)
BEP	0.0270 (0.148)	0.0443 (0.157)	0.0281 (0.150)	0.0474 (0.166)	0.0859 (0.169)	0.0321 (0.159)
N_founders	-0.0327 (0.0443)	-0.0516 (0.0383)	-0.0417 (0.0430)	-0.0383 (0.0462)	-0.0485 (0.0426)	-0.0516 (0.0405)
Start-up size	0.0315** (0.0108)	0.0273* (0.0122)	0.0311** (0.0104)	0.0302** (0.0112)	0.0303** (0.0111)	0.0261* (0.0120)
Market-pull		0.0801 (0.0770)			0.0959 (0.0835)	
Technology-push		0.212** (0.0721)				0.211** (0.0742)
Market Breadth			0.0292 (0.0475)		0.0278 (0.0483)	
Market Depth			0.0281 (0.0463)		0.0336 (0.0467)	
Technical Breadth				0.00399 (0.0380)		-0.00938 (0.0382)
Technical Depth				0.0552 (0.0421)		0.0544 (0.0412)
Constant	0.645*** (0.168)	0.668*** (0.151)	0.472 ^Δ (0.252)	0.533* (0.234)	0.461 ^Δ (0.263)	0.627** (0.219)
inflate bep	-0.450 (0.470)	-0.413 (0.492)	-0.403 (0.472)	-0.425 (0.486)	-0.348 (0.481)	-0.430 (0.502)
Constant	-0.738** (0.283)	-0.831** (0.303)	-0.767** (0.293)	-0.792** (0.301)	-0.817** (0.313)	-0.845** (0.309)
lnalpha						
Constant	-62.22*** (0.114)	-17.25*** (0.120)	-16.26*** (0.509)	-16.05*** (0.401)	-150.0*** (0.109)	-17.65*** (0.403)
Observations	172	172	172	172	172	172
AIC	565.4	561.1	568.0	567.3	568.5	562.6
BIC	590.6	592.6	599.5	598.8	603.1	597.3
Chi2	12.68	24.73	17.08	16.39	14.90	27.36
Log-likelihood	-274.7	-270.6	-274.0	-273.7	-273.2	-270.3
Vuong- AIC	76.96***	8.70***	9.41***	7.20***	8.88***	6.81***
Vuong- BIC	74.54***	5.94***	6.14***	4.49***	7.53***	4.53***
Max VIF	1.21	1.21	1.25	1.39	1.26	1.41

Given the liability of newness associated with start-ups (Stinchcombe, 1965) and the fact that environmental and social goals are not typically primary concerns for digital start-ups, we have opted to use BEP as a variable in the generation of zeros. Nonetheless, we have explored various configurations for the zero-generating process, such as start-up_dimension, start-up_dimension and BEP, and constant term, and results remain consistent with no significant effect on the second component behind the zero-generating process of the model. The same applies to column 6 of Table 5

Standard errors in parentheses ^Δ $p < 0.1$, * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

information are relevant to promote social sustainability⁹. However, as the Vuong test prefers ZINB regression to estimate the model presented in column 6, the Technical Depth turns out to be no more significant¹⁰ (to be compared with Table 13).

All in all, concerning boundary-spanning activities, the most significant actions that seem to influence social sustainability are the engagement in a diverse pool of collaborations (technical and market breadth), more than the exploitation of a few collaborations at the highest level (technical and market depth). Being embedded in a network with different actors may stimulate the exposure of digital start-ups to different perspectives of a variety of stakeholders that may trigger sustainability actions.

Table 6 shows the results when environmental sustainability is the dependent variable.

In column 1, we report the positive and significant effect of the start-up size (the number of employees). Considering the general lack of established routines in start-ups, this result may suggest that new firms with more articulated organizational structures can better orient their environmental actions. In column 2 we register a positive and strongly significant influence of technological strategic orientation, while pointing at a null effect of market strategic orientation (confirming Hp 2). Hypotheses 4a and 4b are not confirmed, given that the coefficients of the technical boundary spanning activities are not statistically significant (see column 4 and 6). This result is sharply different from the one found for social sustainability, indicating that external collaborations seem not among the key drivers to pursuing environmental sustainability, which is a goal that the start-ups reach by exploiting internal resources and practices.

For a visual summary of the result, please see Fig. 2 (where blue arrows indicate supported hypotheses, dotted blue arrows indicate partially supported hypotheses, and orange arrows indicate unsupported hypotheses).

Discussion

With this paper, we contribute to the development of the literature that links digital entrepreneurship and sustainability. First, our study confirms a positive association between strategic orientation and the socio-environmental sustainability of digital start-ups. Second, it shows, in some cases, a positive association between boundary-spanning activities and the socio-environmental sustainability of digital start-ups. More interestingly, our findings confirm that the mechanisms that support the sustainability behavior of digital start-ups are different for social and environmental spheres, suggesting distinctive mechanisms.

All in all, four of our six hypotheses are confirmed, namely Hp1, Hp2, Hp3a and Hp3b.

⁹ As we considered market and technical sharing of information have a high degree of correlation (see Table 4), we decided to test them separately to avoid multicollinearity issues.

¹⁰ As shown in this case, the differences between ZINB and NB regression models estimates suggest care in the selection of the most appropriate count model.

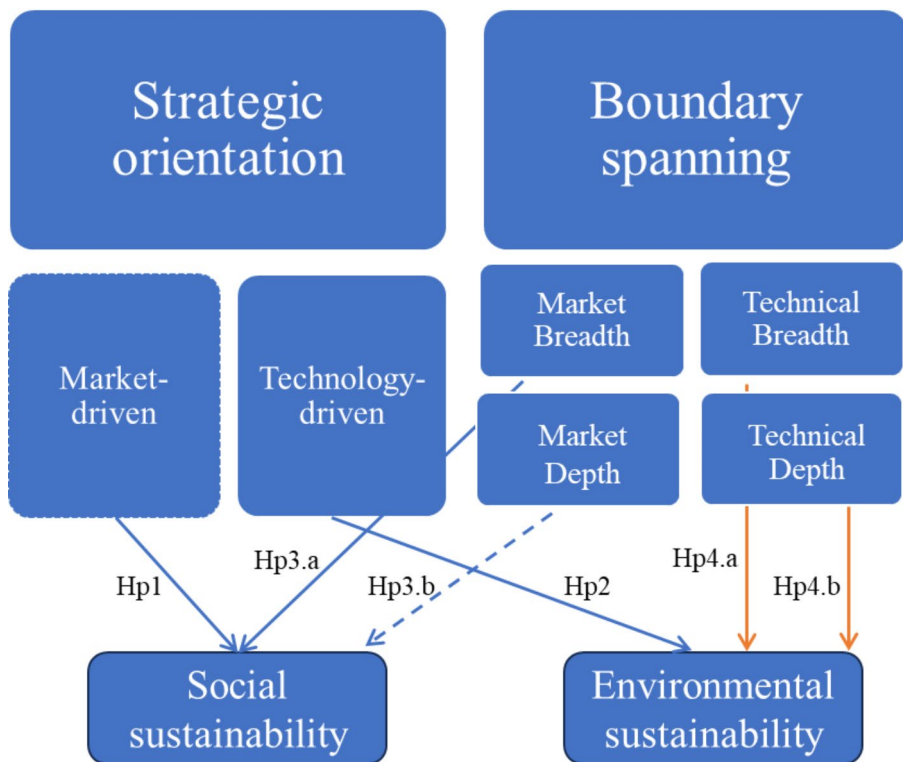


Fig. 2 Results of the hypotheses testing process. Source: authors' elaboration

Concerning strategic orientation, we find that market strategic orientation (market-pull) is relevant only for social sustainability. This result resonates with the literature on social entrepreneurship, underlining the importance of social recognition, branding, and visibility to promote social commitment (De Lange et al., 2017; He et al., 2022). Technological strategic orientation (technology-push) is mainly important for stimulating environmental sustainability, confirming that internal technical capabilities are crucial in developing more complex solutions and promoting initiatives to increase environmental sustainability (Keskin et al., 2013; Kuzma et al., 2020). These two results show a clear distinction path between the two types of sustainability, supporting the importance of analyzing the crucial role of the market and technological strategic orientation for start-ups (in line with Guo et al., 2020).

Concerning boundary spanning, we find that only the exploration of a large portfolio of market collaborations (breadth) is crucial for sustainability, and exclusively for stimulating social sustainability, while the exploitation of a few relationships intensively (depth) leads to more unstable results. Interestingly, both the dimensions of boundary spanning do not influence the environmental sustainability of ICT start-ups.

In the following, we will debate some differences between boundary spanning and the drivers of social and environmental sustainability, respectively.

Social sustainability goals may be better accomplished by a market strategic orientation and using market-related collaborations, which cannot only enhance the start-

up's societal impact but also its reputation in the early stages of its development. In this regard, it is worth noticing that when tested together with market strategic orientation and boundary spanning, the role of more intense relationships seems to be less relevant in comparison to the embeddedness in wider collaboration networks. This result may be explained by the major role played by knowledge diversity and specific expertise in adopting social-related initiatives, which often involve a plethora of actors (Rauter et al., 2019). Finally, we observe a positive and unexpected relationship between more technical boundary spanning and the social sustainability of new ventures. This result opens new venues of research on the role of technical knowledge sharing in general, and of deep relationships with key partners in particular (Rauter et al., 2019; Corić et al., 2020). Future studies can delve into these unexplored mechanisms, analyzing how open innovation paradigm can help the configuration of networks able to generate positive outcomes for societal development (Retolaza et al., 2009; Sedita & Trovò, 2021). Interestingly, we also observe quite similar patterns between technical and market boundary-spanning in promoting social sustainability. This may be interpreted as the general importance of social norms and openness of the context for supporting sustainability practices (Muñoz & Dimov, 2015).

Environmental sustainability goals are strongly influenced by a technological strategic orientation. Interestingly, we notice that collaboration with many different partners is not worthy to pursue environmental sustainability. This may be explained by the stiffness of reaching environmental-related goals, which are often subject to the development of complex technical processes that ask for an accurate partner selection. This is in line with the results by Autio et al. (2021), who pinpointed how business model experimentation is a driver mainly for social sustainability, with less impact on environmental sustainability, which may be more heavily influenced by a coherent internal strategy focused on technical capabilities. Market-oriented companies might be disadvantaged when trying to pursue these goals because there is the risk that their environmental motivations are not deeply rooted in the start-up culture but are driven only by temporary marketing purposes, leading to greenwashing activities that do not have a clear impact on their environmental sustainability.

From the analysis of our control variables, it is interesting to underline peculiar patterns for social and environmental sustainability, which may be worthy of further inquiry.

Gender diversity appears among the determinants of the social sustainability of start-ups (Park et al., 2012; Lammers et al., 2022). This result paves the way for further investigation into the possible consequences of specific hiring policies adopted by start-ups or on the impact of female-dedicated European calls for social innovation projects. Specific analyses on gender diversity among founders may complement this perspective as shown by the study by Colombelli et al. (2024), whose results confirm higher shares of women in social-related start-ups than in green-related start-ups.

Finally, we find that more organized start-ups, which are characterized by a larger number of employees, are more likely to achieve environmental sustainability outcomes since they can also count on internal resources, which are often critical in guaranteeing the integration of external knowledge by enhancing the absorptive capacity of firms. This result is in line with Chen et al. (2015), who also find that “frontline” employees play an important role in shaping the environmental culture

of firms, becoming a precious internal value-added, since they work also as ambassadors of environmental sustainability towards external stakeholders.

Conclusions

This study significantly advances the literature on digital entrepreneurship and sustainability by examining the intricate interplay between strategic orientation, boundary-spanning activities, and socio-environmental sustainability in innovative digital start-ups, particularly ICT ventures based in Italy. Through empirical analysis and hypothesis testing, our research confirms several key insights, shedding light on the distinctive mechanisms driving social and environmental sustainability within these enterprises.

Main limitations

The research has some limitations. First, it is geographically limited to Italy. Although Italy serves as an interesting testbed due to its historical industrial districts, different results may emerge in other European countries, particularly regarding the interactions between start-ups, incumbents, and universities.

Second, the focus on innovative start-ups in the ICT industry. Start-ups listed in the register of innovative companies may exhibit different behaviors and mindsets compared to new ventures not included in the list. However, the availability of a public register to triangulate data with information collected through questionnaires strengthens the reliability of our data.

Third, the reliance on self-reported information from start-ups could introduce potential bias. Nevertheless, this approach provides access to valuable information that would otherwise be impossible to collect.

Implications for scholars

Against the backdrop of the global prevalence of these ventures, driven by the pervasive influence of digital transformation on our economic landscape, our study underscores the importance of nurturing a nascent research strand situated at the crossroads of sustainability and digital entrepreneurship (Gregori & Holzmann, 2020; George et al., 2021; Lammers et al., 2022; Böttcher et al., 2024).

Our findings provide empirical support for the positive association between strategic orientation and the socio-environmental sustainability of digital start-ups. Notably, we demonstrate that market strategic orientation predominantly impacts social sustainability, emphasizing the importance of customer engagement and brand visibility in fostering societal impact. Conversely, technological strategic orientation emerges as a significant driver of environmental sustainability, highlighting the critical role of internal technical capabilities in addressing complex environmental challenges.

Furthermore, our study contributes to the understanding of boundary-spanning activities by revealing their differential effects on social and environmental sustain-

ability. While the breadth of collaborations positively influences social sustainability, the depth of relationships shows less consistent outcomes. This nuanced understanding underscores the importance of diverse knowledge networks in addressing social issues, while also emphasizing the internal focus required for achieving environmental sustainability goals.

However, the inherent resource constraints faced by start-ups often compel them to prioritize short-term objectives, limiting their potential contributions to social and environmental sustainability (Dickel et al., 2018; Hörisch et al., 2019). Hart (1995) lends credence to the notion that the comprehensive integration of sustainability within a firm's strategic framework, encompassing resource allocation and objectives, necessitates a protracted timeline to align the interests of both internal and external stakeholders—a trajectory exemplified by pioneers such as Patagonia. Entrepreneurship, with its inherent promise of reconfiguring established economic paradigms, especially within the digital sphere, assumes a prominent role amidst the continuous proliferation of start-ups on a global scale (Dean & McMullen, 2007; York & Venkataraman, 2010). It is now indisputably evident that socio-environmental sustainability has transcended the realm of discretionary corporate choice; it has evolved into a compelling imperative that will chart our course forward. In the realm of digital entrepreneurship, it represents the fulcrum upon which we can leverage our transformative potential, unveiling a spectrum of sustainable opportunities awaiting exploration. As start-ups navigate this terrain, they will not only shape the dynamics of competition but also contribute significantly to the broader creation of national wealth while fostering a more sustainable future for all (Bassetti et al., 2021).

In addition, the analysis of digital start-ups seen as crucial enablers for solving socio-environmental problems, due to their ownership of digital technologies and capabilities (George et al., 2021; Lammers et al., 2022; Böttcher et al., 2024) may offer a potential solution to the prevailing problem of social detachment associated with the entrepreneurial role models anchored with the Silicon Valley tech lifestyle, still primarily focused on popularity and financial success as main goals (Boyle & Kelly, 2010; Wyrwich et al., 2016).

Implications for entrepreneurs

Practical implications of the study can be found in the identification of specific underlying factors that may support the start-up to invest in green/social policies, accelerating the mental shift towards socio-environmental sustainability and the creation of sustainable dynamic capabilities (Liboni et al., 2022; Allal-Chérif et al., 2023).

In a permanent effort to compete in the market, innovative start-ups have to align their strategic orientation with specific sustainability goals. Those prioritizing social sustainability can benefit from a market-oriented approach focused on customer relationships and brand recognition, while those targeting environmental sustainability should emphasize internal technological capabilities and operational efficiency.

While building a wide portfolio of collaborations is crucial for social sustainability, start-ups should be selective in forming intensive relationships. This ensures access

to diverse knowledge networks while maintaining a focus on internal processes and technological innovation for environmental sustainability. Establishing new relationships has a significant cost for new ventures, and considering the importance of sustainability as a standalone goal, start-ups need to carefully evaluate the purpose of setting these relationships. For start-ups oriented to the market, boundary-spanning may be considered an effective and viable strategy to achieve social sustainability. Conversely, this approach may not be as relevant for start-ups with robust technical capabilities, as their environmental commitment is primarily driven by their internal competencies.

Finally, start-ups should consider the role of gender diversity in promoting social sustainability and explore the implications of specific hiring policies and gender-focused initiatives.

Implications for policymakers

Policymakers play a pivotal role in shaping the regulatory landscape and fostering an environment conducive to socio-environmental sustainability in digital start-ups.

Policymakers should strive to create clear and consistent regulations that encourage sustainable practices in digital start-ups. This includes providing incentives, such as tax breaks or subsidies, for new ventures that prioritize sustainability in their business models. By offering tangible benefits for sustainable initiatives, policymakers can incentivize start-ups to integrate sustainability into their operations. Moreover, specific funding can be allocated for research and development in green technologies, such as renewable energy solutions that might mitigate the extensive energy requirement of digital start-ups (Bettiol, 2023). Finally, policymakers can create platforms and initiatives that facilitate collaboration and knowledge sharing among digital start-ups, as well as between start-ups and other stakeholders, including academic institutions, non-profit organizations, and government agencies. By systematically fostering an ecosystem of collaboration, policymakers can enable start-ups to access diverse expertise and resources to address sustainability challenges (Volkmann et al., 2021).

Further avenues of research

Future research agenda might include an examination of other national settings to see if the drivers of sustainability identified remain valid. In addition, it will be fascinating to see if similar results hold looking at other sectors, also comparing the drivers of innovative start-ups with those of new ventures not included in the list and incumbents.

Appendix

Table 7 Universe and sample regional distribution

Region	Universe		Final sample		Difference
	Frequency	Percentage	Frequency	Percentage	
Abruzzo	52	1.8	2	1.1	-0.7
Basilicata	22	0.8	0	0.0	-0.8
Calabria	65	2.2	5	2.8	0.6
Campania	208	7.1	6	3.4	-3.7
Emilia Romagna	283	9.7	21	11.7	2.0
Friuli Venezia Giulia	49	1.7	3	1.7	0.0
Lazio	344	11.8	26	14.5	2.7
Liguria	54	1.9	4	2.2	0.3
Lombardy	792	27.2	47	26.3	-0.9
Marche	98	3.4	3	1.7	-1.7
Molise	9	0.3	0	0.0	-0.3
Piedmont	155	5.3	10	5.6	0.3
Puglia	127	4.4	6	3.4	-1.0
Sardinia	68	2.3	7	3.9	1.6
Sicily	121	4.2	5	2.8	-1.4
Tuscany	129	4.4	6	3.4	-1.0
Trentino Alto Adige	78	2.7	10	5.6	2.9
Umbria	37	1.3	6	3.4	2.1
Valle D'aosta	7	0.2	0	0.0	-0.2
Veneto	216	7.4	12	6.7	-0.7
Total	2914	100	179	100	0.0

Table 8 Questions used to build our dependent variables

Environmental sustainability orientation items	
What is the company's commitment to the environment?	Possible answer
- It uses production processes with reduced environmental impact	(yes/no)
- suppliers are also assessed from an environmental point of view	(yes/no)
- It carries out training initiatives for personnel on environmental issues	(yes/no)
- It adopts policies to reduce energy consumption	(yes/no)
- It encourages the use of recycled materials	(yes/no)
- It monitors the quantity of waste in relation to production levels	(yes/no)
Social sustainability orientation items	
<i>What is the company's commitment to the society?</i>	<i>Possible answer</i>
- It supports social and environmental projects in the area	(yes/no)
- It carries out communication and community involvement activities (corporate citizenship)	(yes/no)
- It grants liberality in favour of not-for-profit associations, bodies or initiatives	(yes/no)
- It has activated partnerships with non-profit organizations for the realization of a special project	(yes/no)
- It has implemented initiatives to support the enhancement of the territory and cultural heritage	(yes/no)
- It has created a corporate not for profit or promoted voluntary initiatives at the corporate level	(yes/no)

Table 9 Questions used to build our independent variables of interest

Strategic orientation	
What are the main strategic objectives of the company?	Possible answer
- rapid growth in terms of revenues	(yes/no)
- rapid growth in terms of the number of clients	(yes/no)
- product/service development	(yes/no)
- communication development	(yes/no)
- technology development	(yes/no)
- increase the brand awareness	(yes/no)
Market Depth and Breadth	
<i>How deep is the collaboration with the listed partners for each of the following purposes? (INFORMATION ON THE MARKET) (Likert scale from 1=no collaboration to 7=strong collaboration)</i>	<i>Possible answer (values from 1 to 7)</i>
- suppliers	(1,2,3,4,5,6,7)
- clients	(1,2,3,4,5,6,7)
- start-up in the same sector	(1,2,3,4,5,6,7)
- start-up in other sectors	(1,2,3,4,5,6,7)
- companies in the same sector (no start-up)	(1,2,3,4,5,6,7)
- companies in the other sectors (no start-up)	(1,2,3,4,5,6,7)
- private consultants	(1,2,3,4,5,6,7)
- Universities and other research institutions	(1,2,3,4,5,6,7)
Technical Depth and Breadth	
<i>How deep is the collaboration with the listed partners for each of the following purposes? (TECHNICAL INFORMATION) (Likert scale from 1=no collaboration to 7=strong collaboration)</i>	<i>Possible answer (values from 1 to 7)</i>
- suppliers	(1,2,3,4,5,6,7)
- clients	(1,2,3,4,5,6,7)
- start-up in the same sector	(1,2,3,4,5,6,7)
- start-up in other sectors	(1,2,3,4,5,6,7)
- companies in the same sector (no start-up)	(1,2,3,4,5,6,7)
- companies in the other sectors (no start-up)	(1,2,3,4,5,6,7)
- private consultants	(1,2,3,4,5,6,7)
- Universities and other research institutions	(1,2,3,4,5,6,7)

Table 10 Correlation between strategic orientation items

Variables	(1)	(2)	(3)	(4)	(5)	(6)
(1) stra_brand	1.000					
(2) stra_revenues	0.352	1.000				
(3) stra_clients	0.314	0.282	1.000			
(4) stra_prodev	0.326	0.053	0.144	1.000		
(5) stra_comm	0.557	0.284	0.269	0.345	1.000	
(6) stra_tech	0.287	0.096	0.039	0.319	0.261	1.000

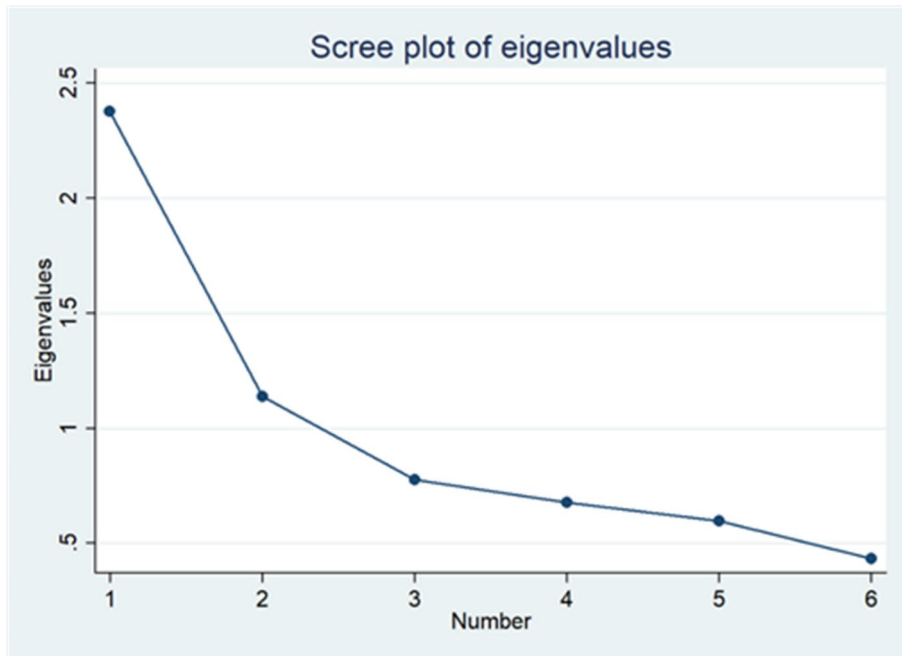


Fig. 3 Scree plot

Table 11 Initial factor analysis with all six items

Factor	Eigenvalue	Difference	Proportion	Cumulative
Factor1	2.377	1.238	0.396	0.396
Factor2	1.139	0.362	0.190	0.586
Factor3	0.776	0.099	0.129	0.715
Factor4	0.678	0.082	0.113	0.828
Factor5	0.595	0.161	0.099	0.928
Factor6	0.434	.	0.072	1.000

LR test: independent vs. saturated: $\chi^2(15) = 169.90$ Prob > $\chi^2 = 0.0000$

Factor loadings (pattern matrix) and unique variances

Variable	Factor1	Factor2	Uniqueness
stra_brand	0.806	0.057	0.347
stra_revenues	0.528	0.542	0.427
stra_clients	0.518	0.505	0.476
stra_prodev	0.575	-0.507	0.412
stra_comm	0.775	-0.021	0.400
stra_tech	0.499	-0.574	0.422

Table 12 Initial factor analysis with all six items – rotated factors

Variable	Label	Factor 1	Factor 2	Uniqueness
stra_brand	increase the brand awareness	0.566	0.414	0.347
stra_revenues	rapid growth in terms of revenues	0.799	-0.180	0.427
stra_clients	rapid growth in terms of the number of clients	0.762	-0.153	0.476
stra_prodev	product/service development	-0.062	0.786	0.412
stra_comm	communication development	0.479	0.466	0.400
stra_tech	technology development	-0.168	0.802	0.422

Table 13 Social sustainability _robustness check

	NB(1)	ZINB (2)	ZINB (3)	ZINB (4)	ZINB (5)	NB(6)
Women	0.727** (0.260)	0.557* (0.216)	0.389* (0.194)	0.499* (0.207)	0.395* (0.199)	0.722** (0.260)
BEP	0.196 (0.167)	0.337* (0.160)	0.309* (0.157)	0.329* (0.151)	0.349* (0.166)	0.113 (0.164)
N_founders	0.0259 (0.0472)	-0.0263 (0.0557)	-0.0336 (0.0746)	-0.0399 (0.0591)	-0.0325 (0.0828)	0.00951 (0.0435)
Start-up size	-0.0203 (0.0377)	0.0460 (0.0434)	0.0257 (0.0863)	0.0370 (0.0638)	0.0236 (0.0917)	-0.0263 (0.0418)
Market-pull		0.106 (0.0965)			0.126 (0.0925)	
Technology-push		0.133^{A°} (0.0707)				0.115 (0.0781)
Market Breadth			0.106* (0.0461)		0.104* (0.0472)	
Market Depth			0.0508 (0.0508)		0.0507 (0.0509)	
Technical Breadth				0.0806* (0.0356)		0.0884* (0.0435)
Technical Depth				0.0627 (0.0382)		0.0899* (0.0443)
Constant	0.206 (0.188)	0.427* (0.179)	-0.136 (0.319)	-0.0553 (0.273)	-0.160 (0.324)	-0.335 (0.275)
inflate bep		0.424 (0.467)	0.613 (0.543)	0.679 (0.537)	0.636 (0.573)	
Constant		-1.057* (0.418)	-1.242* (0.553)	-1.299** (0.474)	-1.304* (0.597)	
lnalpha						
Constant	-0.299 (0.232)	-3.709 (3.662)	-4.712 (13.48)	-14.25*** (3.239)	-4.298 (9.410)	-0.596* (0.294)
Observations	172	172	172	172	172	172
Pseudo R ²	0.010					0.036
AIC	589.3	580.0	574.4	571.8	574.2	580.1
BIC	608.2	611.5	605.9	603.2	608.9	608.4
Chi2	9.517	20.78	18.32	21.63	22.04	27.88
Log-likelihood	-288.7	-280.0	-277.2	-275.9	-276.1	-281.1
LR Test	40.96***					24.43***
Vuong- AIC		0.92	0.85	1.49	0.76	
Vuong- BIC		0.12	0.06	0.14	-0.09	

Standard errors in parentheses ^{A°} $p < 0.1$, * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

Table 14 Environmental sustainability – robustness check

	NB(1)	NB(4)	NB(5)	NB(6)	NB(7)	NB(8)
Women	0.296 (0.276)	0.325 (0.278)	0.270 (0.284)	0.287 (0.277)	0.246 (0.278)	0.348 (0.283)
BEP	0.157 (0.155)	0.156 (0.156)	0.144 (0.154)	0.157 (0.163)	0.188 (0.159)	0.137 (0.159)
N_founders	-0.0415 (0.0461)	-0.0535 (0.0424)	-0.0488 (0.0459)	-0.0527 (0.0471)	-0.0541 (0.0455)	-0.0584 (0.0444)
Start-up size	0.0372* (0.0164)	0.0367* (0.0169)	0.0354* (0.0162)	0.0354* (0.0169)	0.0369* (0.0159)	0.0330^Δ (0.0174)
Market-pull		0.107 (0.0829)			0.131 (0.0843)	
Technology-push		0.213** (0.0826)				0.209* (0.0840)
Market Breadth			0.00864 (0.0499)		0.00184 (0.0505)	
Market Depth			0.0592 (0.0526)		0.0635 (0.0522)	
Technical Breadth				0.00334 (0.0417)		-0.0109 (0.0422)
Technical Depth				0.0911^Δ (0.0484)		0.0861^Δ (0.0487)
Constant	0.254 (0.179)	0.261 (0.170)	0.149 (0.275)	0.125 (0.253)	0.166 (0.281)	0.213 (0.246)
Inalpha						
Constant	-0.737* (0.314)	-0.899* (0.349)	-0.764* (0.321)	-0.826* (0.335)	-0.822* (0.341)	-0.940** (0.363)
Observations	172	172	172	172	172	172
Pseudo R ²	0.007	0.021	0.010	0.014	0.014	0.023
AIC	576.9	573.2	579.5	577.2	579.2	573.9
BIC	595.8	598.4	604.6	602.4	607.6	602.3
Chi2	9.298	20.93	10.48	14.14	12.39	22.72
Log-likelihood	-282.5	-278.6	-281.7	-280.6	-280.6	-278.0
LR Test	19.63***	15.17***	18.88***	16.88***	16.99***	14.08***

Standard errors in parentheses ^Δ $p < 0.1$, * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

Declarations

Competing interests The authors have no competing interests to declare that are relevant to the content of this article.

References

- Allal-Chérif, O., Climent, J. C., & Berenguer, K. J. U. (2023). Born to be sustainable: How to combine strategic disruption, open innovation, and process digitization to create a sustainable business. *Journal of Business Research*, 154, 113379.
- Anand, A., Argade, P., Barkemeyer, R., & Salignac, F. (2021). Trends and patterns in sustainable entrepreneurship research: A bibliometric review and research agenda. *Journal of Business Venturing*, 36(3), 106092.
- Apa, R., & Sedita, S. R. (2017). How (do) internal capabilities and the geography of business networks shape the performance of contractors in public procurement tenders? Evidence from the construction industry. *Construction Management and Economics*, 35(7), 404–419.
- Autio, E., Fu, K., Smit, W., Muftiadi, A., Chiyachantana, C., Prasarnphanich, P., & Park, D. (2021). Adoption of digital technologies, business model innovation, and financial and sustainability performance in startup firms. *Asian Development Bank*, 13(5), 1–40.
- Babic, V., Piccolo, R., & Zarić, S. (2024). What is the purpose of eco-corporate social responsibility in Italian companies? *International Entrepreneurship and Management Journal*, 20, 2611–2628.
- Bassetti, T., Blasi, S., & Sedita, S. R. (2021). The management of sustainable development: A longitudinal analysis of the effects of environmental performance on economic performance. *Business Strategy and the Environment*, 30(1), 21–37.
- Bendig, D., Kleine-Stegemann, L., Schulz, C., & Eckardt, D. (2022). The effect of green start-up investments on incumbents' green innovation output. *Journal of Cleaner Production*, 376, 134316.
- Bettiol, M. (2023). (a cura di). *La sostenibilità Ambientale Del digitale: Il ruolo dei data center*. Padova University.
- Bhupendra, K. V., & Sangle, S. (2015). What drives successful implementation of pollution prevention and cleaner technology strategy? The role of innovative capability. *Journal of Environmental Management*, 155, 184–192.
- Bianchi, M., & Verganti, R. (2021). Entrepreneurs as designers of problems worth solving. *Journal of Business Venturing Design*, 1(1–2), 100006.
- Biswas, A., & Roy, M. (2015). Leveraging factors for sustained green consumption behavior based on consumption value perceptions: testing the structural model. *Journal of Cleaner Production*, 95, 332–340.
- Blevins, D. P., Tsang, E. W., & Spain, S. M. (2015). Count-based research in management: Suggestions for improvement. *Organizational Research Methods*, 18(1), 47–69.
- Boyle, R., & Kelly, L. W. (2010). The celebrity entrepreneur on television: Profile, politics and power. *Celebrity Studies*, 1(3), 334–350.
- Böttcher, T. P., Petry, J., Weking, J., & Hein, A. (2023). Balancing on the Triple-Bottom-Line: Tensions in the Success Factors of Digital Business Models for Sustainability. In 56th Hawaii International Conference on System Sciences (HICSS).
- Böttcher, T. P., Empelmann, S., Weking, J., Hein, A., & Krcmar, H. (2024). Digital sustainable business models: Using digital technology to integrate ecological sustainability into the core of business models. *Information Systems Journal*, 34(3), 736–761.
- Breslin, D. (2008). A review of the evolutionary approach to the study of entrepreneurship. *International Journal of Management Reviews*, 10(4), 399–423.
- Capozza, C., Salomone, S., & Somma, E. (2018). Local industrial structure, agglomeration economies and the creation of innovative start-ups: Evidence from the Italian case. *Entrepreneurship & Regional Development*, 30(7–8), 749–775.
- Chen, J., Chen, Y., & Vanhaverbeke, W. (2011). The influence of scope, depth, and orientation of external technology sources on the innovative performance of Chinese firms. *Technovation*, 31(8), 362–373.

- Chen, Y., Tang, G., Jin, J., Li, J., & Paillé, P. (2015). Linking market orientation and environmental performance: The influence of environmental strategy, employee's environmental involvement, and environmental product quality. *Journal of Business Ethics*, 127(2), 479–500.
- Cohen, W. M., & Levinthal, D. A. (1990). Absorptive capacity: A new perspective on learning and innovation. *Administrative Science Quarterly*, 35(1), 128–152.
- Colombelli, A., Paolucci, E., Raguseo, E., & Elia, G. (2023). The creation of digital innovative start-ups: The role of digital knowledge spillovers and digital skill endowment. *Small Business Economics*, 62(3), 917–937.
- Colombelli, A., D'Ambrosio, A., & Ravetti, C. (2024). Women in innovative start-ups and regional inclusiveness: 'green' and socially-responsible companies. *Regional Studies*, 1–14. <https://doi.org/10.1080/00343404.2024.2340999>
- Ćorić, D. S., Lučić, A., Brečić, R., Šević, A., & Šević, Ž. (2020). An exploration of start-ups' sustainable marketing orientation (SMO). *Industrial Marketing Management*, 91, 176–186.
- De Lange, D. E. (2017). Start-up sustainability: An insurmountable cost or a life-giving investment? *Journal of Cleaner Production*, 156, 838–854.
- Dean, T. J., & McMullen, J. S. (2007). Toward a theory of sustainable entrepreneurship: Reducing environmental degradation through entrepreneurial action. *Journal of Business Venturing*, 22(1), 50–76.
- Denicolai, S., Zucchella, A., & Magnani, G. (2021). Internationalization, digitalization, and sustainability: Are SMEs ready? A survey on synergies and substituting effects among growth paths. *Technological Forecasting and Social Change*, 166, 120650.
- Desmarais, B. A., & Harden, J. J. (2013). Testing for zero inflation in count models: Bias correction for the Vuong test. *The Stata Journal*, 13(4), 810–835.
- Dickel, P., Hörisch, J., & Ritter, T. (2018). Networking for the environment: The impact of environmental orientation on start-ups' networking frequency and network size. *Journal of Cleaner Production*, 179, 308–316.
- Elia, G., Margherita, A., & Passiante, G. (2020). Digital entrepreneurship ecosystem: How digital technologies and collective intelligence are reshaping the entrepreneurial process. *Technological Forecasting and Social Change*, 150, 119791.
- Fabrigar, L. R., Wegener, D. T., MacCallum, R. C., & Strahan, E. J. (1999). Evaluating the use of exploratory factor analysis in psychological research. *Psychological Methods*, 4(3), 272.
- Faludi, J. (2023). How to create social value through digital social innovation? Unlocking the potential of the social value creation of digital start-ups. *Journal of Social Entrepreneurship*, 14(1), 73–90.
- Fraj-Andrés, E., Martínez-Salinas, E., & Matute-Vallejo, J. (2009). Factors affecting corporate environmental strategy in Spanish industrial firms. *Business Strategy and the Environment*, 18(8), 500–514.
- Freeman, J., Carroll, G. R., & Hannan, M. T. (1983). The liability of newness: Age dependence in organizational death rates. *American Sociological Review*, 48(5), 692–710.
- Gatignon, H., & Xuereb, J. M. (1997). Strategic orientation of the firm and new product performance. *Journal of Marketing Research*, 34(1), 77–90.
- George, G., Merrill, R. K., & Schillebeeckx, S. J. (2021). Digital sustainability and entrepreneurship: How digital innovations are helping tackle climate change and sustainable development. *Entrepreneurship Theory and Practice*, 45(5), 999–1027.
- González-Moreno, Á., Triguero, Á., & Sáez-Martínez, F. J. (2019). Many or trusted partners for eco-innovation? The influence of breadth and depth of firms' knowledge network in the food sector. *Technological Forecasting and Social Change*, 147, 51–62.
- Greco, M., Grimaldi, M., & Cricelli, L. (2016). An analysis of the open innovation effect on firm performance. *European Management Journal*, 34(5), 501–516.
- Greene, M. (2003). *Econometric Analysis*. Prentice Hall.
- Gregori, P., & Holzmann, P. (2020). Digital sustainable entrepreneurship: A business model perspective on embedding digital technologies for social and environmental value creation. *Journal of Cleaner Production*, 272, 122817.
- Gu, Y., & Su, D. (2018). Innovation orientations, external partnerships, and start-ups' performance of low-carbon ventures. *Journal of Cleaner Production*, 194, 69–77.
- Guo, H., Wang, C., Su, Z., & Wang, D. (2020). Technology push or market pull? Strategic orientation in business model design and digital start-up performance. *Journal of Product Innovation Management*, 37(4), 352–372.
- Guo, H., Li, X., & Wang, C. (2023). When the window of opportunity opens: How does open search impact the business model design of digital start-ups? *Asia Pacific Business Review*, 29(4), 852–875.

- Hair, J. F., Black, W. C., Babin, B. J., Anderson, R. E., & Tatham, R. L. (2014). *Multivariate data analysis: Global edition*.
- Hakala, H. (2011). Strategic orientations in management literature: Three approaches to understanding the interaction between market, technology, entrepreneurial and learning orientations. *International Journal of Management Reviews*, 13(2), 199–217.
- Hart, S. L. (1995). A natural-resource-based view of the firm. *Academy of Management Review*, 20(4), 986–1014.
- He, T., Liu, M. J., Phang, C. W., & Luo, J. (2022). Toward social enterprise sustainability: The role of digital hybridity. *Technological Forecasting and Social Change*, 175, 121360.
- Holmes, S., & Smart, P. (2009). Exploring open innovation practice in firm-nonprofit engagements: A corporate social responsibility perspective. *R&D Management*, 39(4), 394–409.
- Hörisch, J., Kollat, J., & Brieger, S. A. (2019). Environmental orientation among nascent and established entrepreneurs: An empirical analysis of differences and their causes. *International Journal of Entrepreneurial Venturing*, 11(4), 373–393.
- Horne, J., & Fichter, K. (2022). Growing for sustainability: Enablers for the growth of impact startups—A conceptual framework, taxonomy, and systematic literature review. *Journal of Cleaner Production*, 349, 131163.
- Jansson, J., Nilsson, J., Modig, F., & Hed Vall, G. (2017). Commitment to sustainability in small and medium-sized enterprises: The influence of strategic orientations and management values. *Business Strategy and the Environment*, 26(1), 69–83.
- Jesus, G. M. K., & Jugend, D. (2021). How can open innovation contribute to circular economy adoption? Insights from a literature review. *European Journal of Innovation Management*, 26(1), 65–98.
- Kallinikos, J., Aaltonen, A., & Marton, A. (2013). The ambivalent ontology of digital artifacts. *MIS Quarterly*, 37(2), 357–370.
- Keskin, D., Diehl, J. C., & Molenaar, N. (2013). Innovation process of new ventures driven by sustainability. *Journal of Cleaner Production*, 45, 50–60.
- Kuzma, E., Padilha, L. S., Sehnem, S., Julkovski, D. J., & Roman, D. J. (2020). The relationship between innovation and sustainability: A meta-analytic study. *Journal of Cleaner Production*, 259, 120745.
- Lammers, T., Rashid, L., Kratzer, J., & Voinov, A. (2022). An analysis of the sustainability goals of digital technology start-ups in Berlin. *Technological Forecasting and Social Change*, 185, 122096.
- Laroche, M., Bergeron, J., & Barbaro-Forleo, G. (2001). Targeting consumers who are willing to pay more for environmentally friendly products. *Journal of Consumer Marketing*, 18(6), 503–520.
- Laursen, K., & Salter, A. (2006). Open for innovation: The role of openness in explaining innovation performance among UK manufacturing firms. *Strategic Management Journal*, 27(2), 131–150.
- Lenton, T. M., Held, H., Kriegler, E., Hall, J. W., Lucht, W., Rahmstorf, S., & Schellnhuber, H. J. (2008). Tipping elements in the Earth's climate system. *Proceedings of the National Academy of Sciences*, 105(6), 1786–1793.
- Liboni, L. B., Cezarino, L. O., Alves, M. F. R., Jabbour, C., C. J., & Venkatesh, V. G. (2022). Translating the environmental orientation of firms into sustainable outcomes: The role of sustainable dynamic capability. *Review of Managerial Science*, 17(4), 1125–1146.
- Lopez-Vega, H., Tell, F., & Vanhaverbeke, W. (2016). Where and how to search? Search paths in open innovation. *Research Policy*, 45(1), 125–136.
- Liu, B., Lu, L., Zhang, H., & Liu, C. (2021). Strategic choices for social responsibility of start-ups in China. *Frontiers in Psychology*, 12, 719454.
- Lombardi, M., & Mazzoni, L. (2023). Innovation policies for the emerging cyber-physical world: A research agenda. *Technology Analysis & Strategic Management*, 36(10), 3020–3033.
- Mazzoni, L., Lazzaretto, L., & Innocenti, N. (2021). Entrepreneurship, complexity and the emergent order in the techno-economic scenario of the twenty-first century. Evidence from a field study in Tuscany. *Industry and Innovation*, 28(5), 570–593.
- Mazzoni, L., & Innocenti, N. (2024). What conditions favor high-potential entrepreneurship? Unpacking the nexus between the industrial structure and startup typologies. *Small Business Economics*, 62(3), 1201–1222.
- Michelino, F., Cammarano, A., Celone, A., & Caputo, M. (2019). The linkage between sustainability and innovation performance in IT hardware sector. *Sustainability*, 11(16), 4275.
- Morgan, R. E., & Strong, C. A. (1998). Market orientation and dimensions of strategic orientation. *European Journal of Marketing*, 32(11/12), 1051–1073.

- Mu, J., Thomas, E., Peng, G., & Di Benedetto, A. (2017). Strategic orientation and new product development performance: The role of networking capability and networking ability. *Industrial Marketing Management*, 64, 187–201.
- Mulligan, C., & Kelly, F. (2021). Digital Entrepreneurship: Ensuring True Compliance with Sustainable Development Goals (SDGs). In 2021 IEEE International Conference on Technology and Entrepreneurship (ICTE) (pp. 1–7). IEEE.
- Muñoz, P., & Dimov, D. (2015). The call of the whole in understanding the development of sustainable ventures. *Journal of Business Venturing*, 30(4), 632–654.
- Nadjahi, C., Louahlia, H., & Lemasson, S. (2018). A review of thermal management and innovative cooling strategies for data center. *Sustainable Computing: Informatics and Systems*, 19, 14–28.
- Nambisan, S., Wright, M., & Feldman, M. (2019). The digital transformation of innovation and entrepreneurship: Progress, challenges and key themes. *Research Policy*, 48(8), 103773.
- Narver, J. C., & Slater, S. F. (1990). The effect of a market orientation on business profitability. *Journal of Marketing*, 54(4), 20–35.
- Noble, C. H., Sinha, R. K., & Kumar, A. (2002). Market orientation and alternative strategic orientations: A longitudinal assessment of performance implications. *Journal of Marketing*, 66(4), 25–39.
- Park, S. J., Choi, S., & Kim, E. J. (2012). The relationships between socio-demographic variables and concerns about environmental sustainability. *Corporate Social Responsibility and Environmental Management*, 19(6), 343–354.
- Porter, M. E., & Heppelmann, J. E. (2015). How smart, connected products are transforming companies. *Harvard Business Review*, 93(10), 96–114.
- Powell, W. W., Koput, K. W., & Smith-Doerr, L. (1996). Interorganizational collaboration and the locus of innovation: Networks of learning in biotechnology. *Administrative Science Quarterly*, 41(1), 116–145.
- Rauter, R., Globocnik, D., Perl-Vorbach, E., & Baumgartner, R. J. (2019). Open innovation and its effects on economic and sustainability innovation performance. *Journal of Innovation & Knowledge*, 4(4), 226–233.
- Retolaza, J. L., Ruiz, M., & San-Jose, L. (2009). CSR in business start-ups: An application method for stakeholder engagement. *Corporate Social Responsibility and Environmental Management*, 16(6), 324–336.
- Riandita, A., Brostro, m, A., Cagliano, R., & Feldmann, A. (2019). The Journey of Sustainability Start-ups through Establishing Major Alliance Partners. In Academy of Management Proceedings, 2019(1), Article 19178.
- Rosenkopf, L., & Nerkar, A. (2001). Beyond local research: boundary-spanning, exploration, and impact in the optical disk industry. *Strategic Management Journal*, 22(4), 287–306.
- Scheffer, M., Carpenter, S. R., Lenton, T. M., Bascompte, J., Brock, W., Dakos, V., & Vandermeer, J. (2012). Anticipating critical transitions. *Science*, 338(6105), 344–348.
- Schotter, A. P., Mudambi, R., Doz, Y. L., & Gaur, A. (2017). Boundary spanning in global organizations. *Journal of Management Studies*, 54(4), 403–421.
- Sedita, S. R., & Apa, R. (2016). In M. D. Parrilli, R. D. Fitjar, & A. Rodriguez-Pose (Eds.), *The role of Open Innovation-oriented strategies in the Innovation performance of Mechanical Engineering Start-Up firms in Northern Italy* (pp. 142–159). Innovation drivers and regional innovation strategy.
- Sedita, S. R., Apa, R., Bassetti, T., & Grandinetti, R. (2019). Incubation matters: Measuring the effect of business incubators on the innovation performance of start-ups. *R&D Management*, 49(4), 439–454.
- Sedita, S. R., & Trovò, A. (2021). Social Entrepreneurship: Determinants of action in Italy. In S. R. Sedita, & S. Blasi (Eds.), *Rethinking clusters: Place-based Value Creation in sustainability transitions* (pp. 175–193). Springer International Publishing.
- Seuring, S., & Gold, S. (2013). Sustainability management beyond corporate boundaries: From stakeholders to performance. *Journal of Cleaner Production*, 56, 1–6.
- Shane, S., & Venkataraman, S. (2000). The promise of entrepreneurship as a field of research. *Academy of Management Review*, 25(1), 217–226.
- Spender, J. C., Corvello, V., Grimaldi, M., & Rippa, P. (2017). Start-ups and open innovation: A review of the literature. *European Journal of Innovation Management*, 20(1), 4–30.
- Stinchcombe, A. L. (1965). Social structure and organizations. In J. C. March (Ed.), *Handbook of organizations* (pp. 142–193). Rand McNally.
- Straughan, R. D., & Roberts, J. A. (1999). Environmental segmentation alternatives: A look at green consumer behavior in the new millennium. *Journal of Consumer Marketing*, 16(6), 558–575.

- Testa, F., Gusmerottia, N. M., Corsini, F., Passetti, E., & Iraldo, F. (2016). Factors affecting environmental management by small and micro firms: The importance of entrepreneurs' attitudes and environmental investment. *Corporate Social Responsibility and Environmental Management*, 23(6), 373–385.
- Volkmann, C., Fichter, K., Klofsten, M., & Audretsch, D. B. (2021). Sustainable entrepreneurial ecosystems: An emerging field of research. *Small Business Economics*, 56(3), 1047–1055.
- von Briel, F., Davidsson, P., & Recker, J. (2018). Digital technologies as external enablers of new venture creation in the IT hardware sector. *Entrepreneurship Theory and Practice*, 42(1), 47–69.
- Wang, Y., Shi, S., Chen, Y., & Gursay, D. (2019). An examination of market orientation and environmental marketing strategy: The case of Chinese firms. *The Service Industries Journal*, 39(15–16), 1046–1071.
- Wang, J., Xue, Y., & Yang, J. (2020). Boundary-spanning search and firms' green innovation: The moderating role of resource orchestration capability. *Business Strategy and the Environment*, 29(2), 361–374.
- Wang, D., Song, J., Sun, X., & Wang, X. (2022). A study on the impact of boundary-spanning search on the sustainable development performance of technology start-ups. *Sustainability*, 14(15), 9182.
- Wyrwich, M., Stuetzer, M., & Sternberg, R. (2016). Entrepreneurial role models, fear of failure, and institutional approval of entrepreneurship: A tale of two regions. *Small Business Economics*, 46(3), 467–492.
- Yoo, Y., Boland Jr, R. J., Lyytinen, K., & Majchrzak, A. (2012). Organizing for innovation in the digitized world. *Organization Science*, 23(5), 1398–1408.
- York, J. G., & Venkataraman, S. (2010). The entrepreneur–environment nexus: Uncertainty, innovation, and allocation. *Journal of Business Venturing*, 25(5), 449–463.
- Yunxiong Li, G., Ascani, A., & Iammarino, S. (2022). The material basis of modern technologies. A case study on rare metals. Working Papers 59; Birkbeck Centre for Innovation Management: London, UK; Birkbeck, University of London: London, UK, 2022; Available online: <https://eprints.bbk.ac.uk/id/eprint/47608/>
- Zaheer, H., Breyer, Y., & Dumay, J. (2019). Digital entrepreneurship: An interdisciplinary structured literature review and research agenda. *Technological Forecasting and Social Change*, 148, 119735.

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